MIL-R-15189B
9 September 1986
SUPERSEDING
MIL-R-15189A
12 December 1951
(See 6.4)

MILITARY SPECIFICATION

RESIN, ALKYD PHENOLIC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a phenolic resin for use in the manufacture of a mixing varnish employed in the preparation of coatings.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications. The following specifications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and the supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

TT-L-1155 - Linseed Oil, Alkali Refined.
TT-R-266 - Resin, Alkyd; Solutions.
TT-T-291 - Thinner, Paint, Mineral Spirits, Regular and Odorless.

(Copies of specifications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING MATERIALS (ASTM)

D 12 - Standard Specification for Raw Tung Oil.
D 154 - Standard Methods of Testing Varnishes.
D 1475 - Standard Test Method for Density of Paint, Varnish, Lacquer, and Related Products. (DoD adopted)
D 1542 - Standard Method of Qualitative Tests for Rosin in Varnishes. (DoD adopted)
D 1544 - Standard Test Method for Color of Transparent Liquids (Gardner Color Scale). (DoD adopted)
D 1545 - Standard Test Method for Viscosity of Transparent Liquids by Bubble Time Method. (DoD adopted)
D 1640 - Standard Test Methods for Drying, Curing, or Film Formation of Organic Coatings at Room Temperature.
D 1642 - Standard Test Methods for Elasticity or Toughness of Varnishes. (DoD adopted)
D 1643 - Standard Test Methods for Gas Checking and Draft Test of Varnish Films. (DoD adopted)
D 1647 - Standard Test Methods for Resistance of Dried Films of Varnishes to Water and Alkali.
D 1951 - Standard Test Method for Ash in Drying Oils and Fatty Acids. (DoD adopted)
D 2369 - Standard Test Method for Volatile Content of Coatings. (DoD adopted)
E 28 - Standard Test Method for Softening Point by Ring-and-Ball Apparatus. (DoD adopted)

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103).

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services).

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Materials. The phenol-formaldehyde resin used in making the final product shall be made from aliphatic para-substituted phenols with substituting groups containing four to eight carbon atoms. The resin shall be a dry granular material.

3.2 Physical characteristics. When tested as specified in 4.3, the resin, and a varnish prepared as specified in 4.3.3, shall conform to the physical characteristics requirements of table I.

TABLE I. Physical characteristics.

Component or property	Minimum	Maximum
For the resin		
Specific gravity at 25°C (77°F)	1.00	1.10
Softening range (°C)	120	140
Volatile content at 300°C (572°F) (percent mass)	—	5
Ash (percent mass)	—	0.2
Rosin and rosin derivatives 1/	—	—
For the varnish		
Nonvolatile content (percent mass)	59	61
Density		
Kilograms per liter	0.90	0.92
Pounds per gallon	(7.5)	(7.7)
Viscosity, Gardner-Holt	L	N
Color, Gardner Color Standard	—	15
Drying time		
Set to touch (hours)	0.5	2
Dry hard (hours)	—	12
Compatibility with alkyd resin 2/	—	—
Toughness (percent)	90	—
Gas checking 3/	—	—
Water resistance 4/	—	—
Alkali resistence 5/	—	—
Appearance 6/	—	—
Skinning 7/	—	—

1/ No evidence of rosin and rosin derivatives.
2/ No evidence of incompatibility.
3/ No evidence of frosting, checking, wrinkling, crow's-footing, or any other defect.
4/ No evidence of blistering and no more than very slight whitening, dulling, checking or other signs of deterioration when compared to an unimmersed panel after a 2-hour recovery period.
5/ No noticeable discoloration or change in appearance after 6-hours immersion, rinsing, and 2-hour recovery. (Very light hazing that can be removed by hand buffing is permissible and effects only at the bottom of the test tube may be disregarded.)

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6/ No evidence of sediment or suspended matter during shaking or after settling for 24 hours.
7/ No evidence of skinning after 72 hours.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Sampling. Two samples shall be taken at random from different packages from each lot, batch, day's pack, or other unit of production in a shipment. When no markings distinguishing between units of production appear, samples shall be taken from different packages in the ratio of two samples for each (10,000 pounds) (4,540 kilograms (kg)), except that for shipments of less than 10,000 pounds two samples shall be taken. The samples may be tested separately or as a composite sample, after blending in equal quantities the samples from the same production unit.

4.3 Test procedures. Tests shall be conducted in accordance with table II and as specified in 4.3.1 through 4.3.4.

4.3.1 Specific gravity. Suspend on a loop of fine wire and weigh to the nearest milligram (using a balance sensitives to the nearest 0.1 milligram) a piece of resin 4 to 6 grams in weight, first in air and then in water at 25 degrees Celsius (°C) (77 degrees Fahrenheit (°F)). The resin shall be free from bubbles, cracks, dust, and abrupt depressions. Divide the weight of the resin (without wire) by the difference in the weighings (in air and in water) to obtain the specific gravity. Repeat the determination with three separate pieces of resin and report the average result.

Formula:

$$\text{Specific gravity} = \frac{\text{weight of resin}}{\text{weight in air minus weight in water}}$$

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4.3.2 Volatile content. Crush a sample of resin to obtain particles no more than 3 millimeters (mm) in diameter. Weigh to the nearest milligram a 100 milliliter (mL) beaker before and after adding 10 grams of the crushed resin. Place the beaker with sample into a muffle furnace or oven at 300°C (572°F), maintain it at this temperature for 15 minutes, and remove, cool, and weigh it. Report the percent loss in weight of the sample as volatile matter.

TABLE II. Test procedures.

Characteristics	Applicable ASTM test method	Test paragraph
For the resin		
Specific gravity at 25°C (77°F)	---	4.3.1
Softening range (°C)	E 28	---
Volatile content at 300°C (572°F) (percent mass)	---	4.3.2
Ash (percent mass)	1/ D 1951	---
Rosin and rosin derivatives	D 1542	---
For the varnish		
Nonvolatile content (percent mass)	D 2369, procedure B	4.3.3
Density	D 1475	---
Viscosity, Gardner-Holt	D 1545	---
Color, Gardner Color Standard	D 1544	---
Drying time	D 1640	---
Compatibility with alkyd resin	---	4.3.4
Toughness (percent)	D 1642, method A	---
Gas checking	D 1643, method B	---
Water resistance	D 1647	---
Alkali resistance	D 1647	---
Appearance	D 154, section 5	---
Skinning	D 154, section 6	---

1/ Instead of 100 grams of oil, use 5 grams of resin.

4.3.3 Varnish preparation. Prepare a test varnish by suitably cooking the following ingredients in the amounts or proportions shown (but adjust the amount of thinner to obtain the required nonvolatile content):

700 grams	Resin
679 grams	Alkali refined linseed oil (TT-L-1155)
687 grams	Tung oil (China wood), raw (ASTM D 12)
1407 grams	Paint thinner (TT-T-291, type II)
as required	Non lead driers

4.3.4 Compatibility with alkyd resin. Place a 50-mL sample of the test varnish (see 4.3.3) into a clean, dry, glass-stoppered, 100-mL graduated cylinder. Add 50-mL alkyd resin solution (TT-R-266, type I, class A) and agitate until well mixed. Examine for cloudiness, precipitation, or other signs of incompatibility, and reexamine after allowing to stand for 24 hours.

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4.4 Inspection of packaging. Sample packages and packs, and the inspection of the preservation-packaging, packing and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

5. PACKAGING

5.1 Packaging, packing and marking. Packaging, packing and marking of the resin shall be as specified (see 6.2).

6. NOTES

6.1 Intended use. This specification is for a resin to be used in the manufacture of Government specification coatings and is not intended for direct acquisition by the Government.

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number and date of this specification.
(b) Packaging, packing and marking required (see 5.1).

6.3 Subject term (key word) listing.

Alkyd
Coatings
Phenolic
Resin
Varnish

6.4 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Navy - SH

Preparing activity:
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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

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1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-R-15189B	RESIN, ALKYD PHENOLIC

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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